Project Name: Inventory Tracker

Team Members:

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Problem (The "Why")

- **Current Issue:** Asset discrepancies (*ghost assets*) are a common challenge in asset management, occurring when recorded asset information does not match the actual physical assets (lost, broken, or untracked). Furthermore, asset identification relies on passive label stickers, forcing auditors to perform manual, slow, and error-prone alphanumeric text searches during audits.
- **Impact:** Wasted spending on software licenses, redundant procurement for untracked or non-existent assets.

Proposed Solution

- **Our Approach:** Develop a centralized web based asset tracking system where every asset is assigned a unique QR Code. Instead of manually searching for alphanumeric asset IDs, users can scan the QR Code to instantly access asset information, update its status, verify its location, and record audit activities.
- **Key Value:** The system simplifies asset identification, reduces manual verification, improves audit efficiency, and provides accurate, real-time visibility of company assets.

Core Features

- **Feature 1:** Asset Management
  Provide a centralized system to register, update, and manage all company assets. Each asset stores detailed information such as asset ID, name, category, location, assigned user, purchase date, and current status. The

system serves as the single source of truth for asset records, reducing inconsistencies and simplifying asset administration.

- **Feature 2:** QR Code Based Asset Identification
  Automatically generate a unique QR Code for every registered asset. Instead of manually searching and verifying alphanumeric asset IDs, users can scan the QR Code to instantly retrieve the asset's information. This significantly speeds up asset identification and minimizes human error during daily operations and audits.

- **Feature 3:** Asset Audit & Verification
  Enable auditors to perform asset verification by scanning the asset's QR Code and confirming its current condition, location, and status. Every audit activity is recorded with timestamps and verification results, creating a complete audit history. This feature helps maintain accurate asset records, improves audit efficiency, and reduces the risk of asset discrepancies (*ghost assets*).

High-Level Architecture & Quality Assurance

- **Architecture Pattern:** Clean Architecture
- **Design Patterns Used:** Repository Pattern, Dependency Injection
- **Code Quality Goal:** Keep CI green, write unit tests for core logic. Test each service's API automatically before merging to catch breaking changes early.

Identified Challenges & Risks

- **Risk 1:** Preventing Data Conflicts During Simultaneous Updates.
- **Risk 2:** Isolating the Third-Party QR Code Library to Maintain Easy Unit Testing